



Instituto Tecnológico de Costa Rica

Escuela de Ingeniería en Computadores

Modelación Hardware/Software con Orientación a Objetos • CE-5507

Project 3 - Design Document

Interactive product with hardware and software

Smart Pill Reminder Box

Object-Oriented pill reminder system with a software app and hardware interface

System Design • ESP32 • Pill Reminder
C • C++ • Arduino IDE • Web Design

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Abstract

This document presents the design proposal for the Smart Pill Reminder Box, an intelligent medication reminder device developed as part of Project 3 for the CE-5507 Hardware/Software Modeling with Object-Oriented Design course at the Instituto Tecnológico de Costa Rica. The system is designed to assist older adults with mild to moderate cognitive impairment, particularly those diagnosed with Alzheimer’s disease or related conditions, in correctly following their medication routines.

The proposed solution integrates an ESP32 microcontroller with a set of input and output components, including magnetic reed switches, drawer LEDs, a buzzer, and an LCD screen. The device communicates wirelessly with a mobile application used by caregivers to configure medication schedules, assign medications to physical drawers, and receive feedback regarding patient interaction with the device.

The system architecture follows an object-oriented design organized into four main layers: the mobile application, the domain model, the device services running on the ESP32, and the hardware abstraction layer. Communication between the mobile application and the physical device is handled through the MQTT protocol over WiFi.

This document describes the use case and class diagrams, the system architecture and communication flow, the design decisions that guided the development of both the hardware and software components, the physical component layout, and the materials selected for the construction of the prototype.

Keywords

Object-oriented programming, embedded systems, ESP32, medication reminder, older adults, cognitive impairment, Alzheimer’s disease, mobile application, MQTT, hardware/software integration, accessible design, prototype.

1. Introduction

Medication adherence is a critical factor in the effectiveness of medical treatments, particularly for older adults who must follow complex and time-sensitive schedules involving multiple medications [1]. For individuals affected by conditions such as Alzheimer’s disease or other forms of cognitive impairment, the challenge of remembering when and how to take each medication can become a significant health risk. Missed doses, duplicated intake, or incorrect administration directly affect treatment outcomes and can lead to serious consequences for the patient’s physical and mental health [1].

Many older adults with cognitive impairment also experience a progressive loss of autonomy, which increases their dependence on caregivers and family members for the management of daily routines [2]. While this support is necessary, constant dependence can also generate feelings of frustration and loss of independence. There is therefore a need for assistive solutions that support medication adherence while preserving a meaningful degree of user autonomy [3].

The Smart Pill Reminder Box addresses this need by combining a physical pill organizer with an intelligent reminder system. The device organizes medications into labeled compartments and activates visual, auditory, and textual alerts when a scheduled dose is due [4]. The correct drawer is identified through blinking LEDs and an LCD screen displaying the medication name and compartment number. A magnetic reed switch installed in each drawer detects whether it has been opened and closed, allowing the system to confirm that the user interacted with the correct compartment [5].

The device is configured and monitored through a mobile application designed for caregivers. From the application, the caregiver can register medications, assign them to specific drawers, define reminder schedules, and receive notifications when a dose is missed or when medication supplies are running low. The communication between the mobile application and the ESP32 microcontroller is handled wirelessly using the MQTT protocol over WiFi, a lightweight publish-subscribe messaging standard widely adopted in IoT healthcare systems [6].

The remainder of this document is structured as follows. Section II presents the use case diagram, describing the main interactions between the system actors. Section III presents the state diagram, modeling the internal behavior of the ESP32 firmware throughout the reminder lifecycle. Section IV presents the class diagram, detailing the object-oriented structure of both the mobile application and the embedded software. Section V describes the system architecture, including the hardware and software communication flow, the main architectural blocks, and the overall interaction sequence. Section VI outlines the design decisions that guided the development of both the hardware and software components. Section VII describes the physical component layout and distribution within the device. Section VIII presents the materials selected for the construction of the prototype and the rationale behind each choice. Finally, Section IX presents the conclusions drawn from the design process.

2. Use Case Diagram

The following use case diagram represents the main interactions between the users, the mobile application, and the Smart Pill Reminder Box system. It illustrates how caregivers configure medication schedules and how older adults interact with the device during the reminder process. The diagram also shows how the system provides visual and auditory feedback, monitors drawer interactions, and communicates relevant information back to the mobile application to support medication adherence and patient monitoring.

Use case diagrams provide a clear overview of the system's functionalities and the relationships between users and system components. By visually mapping these interactions, it becomes easier to identify the main system requirements, define the responsibilities of each actor, and detect possible limitations or interaction issues during the early stages of development. This contributes to a more organized design process and helps ensure that both the hardware and software components respond effectively to the needs of the users.

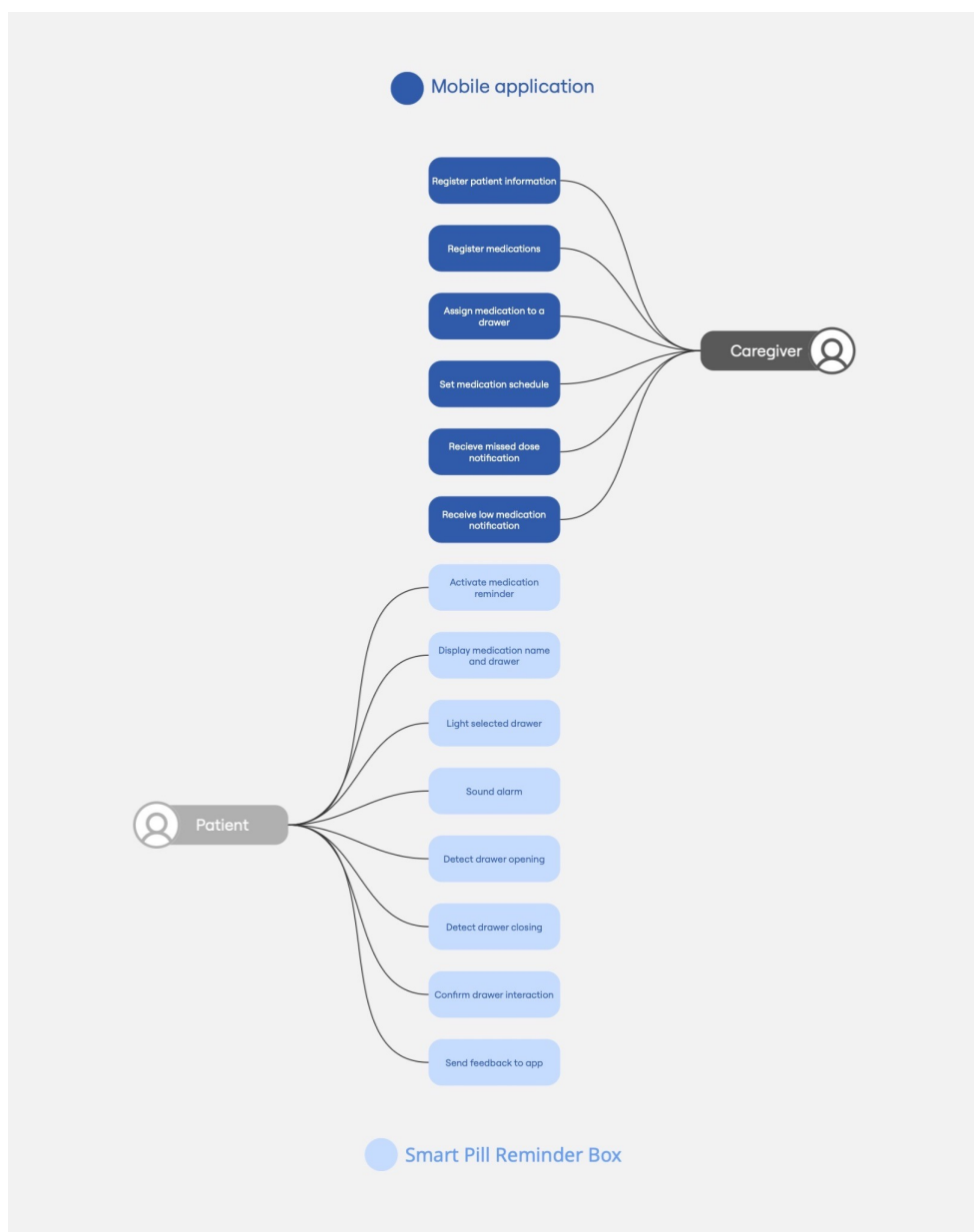


Figure 1: Use case diagram of the Smart Pill Reminder Box system

3. State Diagram

The following state diagram models the internal behavior of the ESP32 firmware throughout the full medication reminder lifecycle. Each state represents a distinct operational mode of the system, and each transition is triggered by a detectable event from the hardware inputs, the internal clock, or the communication layer.

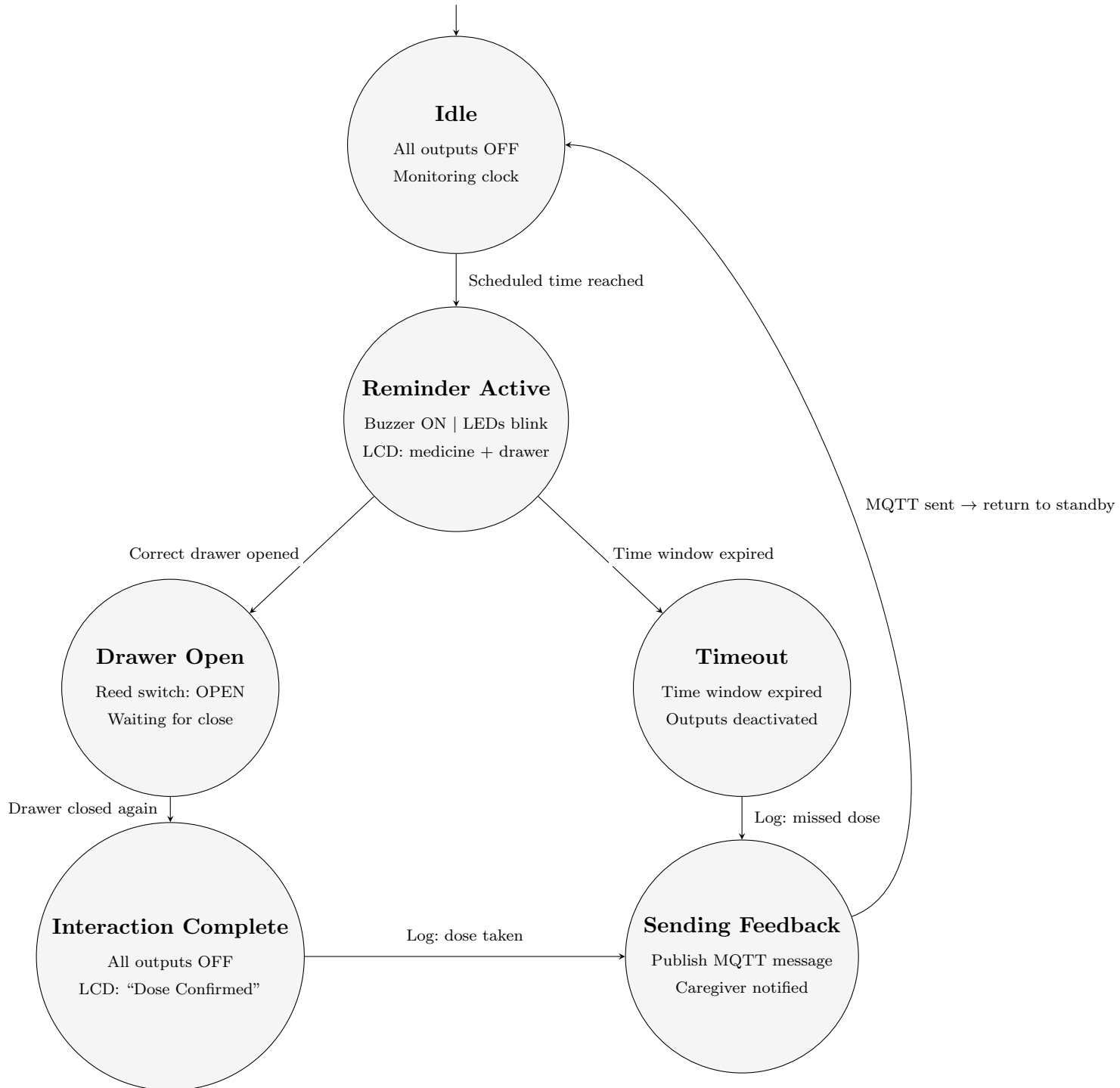


Figure 2: State diagram of the Smart Pill Reminder Box ESP32 firmware

4. Classes Diagram

Here's the class diagram for our solution:

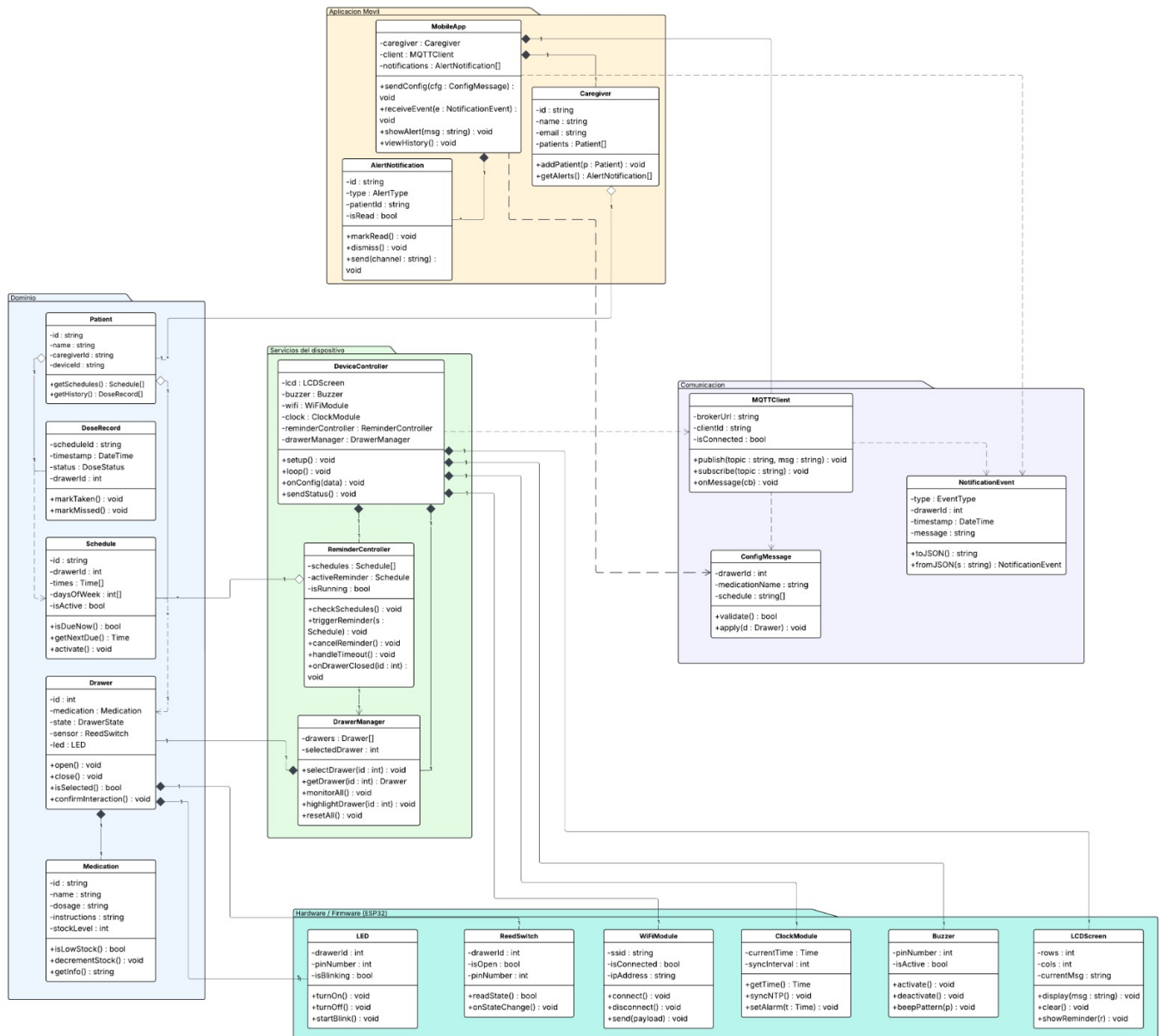


Figure 3: Classes diagram of the Smart Pill Reminder Box system

The class diagram is organized into four main layers that reflect the separation of responsibilities within the system.

The **Mobile Application** layer contains the **MobileApp**, **Caregiver**, and **AlertNotification** classes. The **MobileApp** class manages the caregiver session and handles incoming notification events, while **AlertNotification** represents the alerts sent to the caregiver when a dose is missed or completed.

The **Domain** layer defines the core data entities of the system: **Patient**, **Medication**, **Schedule**, **Drawer**, and **DoseRecord**. These classes represent the business logic of the application, including medication assignments, reminder schedules, and interaction history.

The **Device Services** layer contains the classes running on the ESP32: `DeviceController`, `ReminderController`, and `DrawerManager`. The `DeviceController` acts as the main entry point of the firmware, coordinating the reminder logic through `ReminderController` and managing drawer states through `DrawerManager`.

The **Communication** layer includes `MQTTClient`, `ConfigMessage`, and `NotificationEvent`, which handle the wireless data exchange between the ESP32 and the mobile application using the MQTT protocol.

Finally, the **Hardware/Firmware** layer at the bottom represents the physical components abstracted as classes: `LED`, `ReedSwitch`, `WiFiModule`, `ClockModule`, `Buzzer`, and `LCDScreen`, each encapsulating the control logic for its corresponding hardware element.

5. System Architecture

5.1. Hardware and Software Communication

The system architecture is based on the interaction between a mobile application and an ESP32 microcontroller. The mobile application is responsible for sending the reminder configuration, while the ESP32 is responsible for controlling the physical behavior of the pill reminder box.

The communication between the mobile application and the ESP32 is wireless, using the connectivity capabilities of the ESP32. Through this communication, the application sends the medicine name, assigned drawer number, and scheduled reminder time to the ESP32.

The communication flow is bidirectional. The mobile application sends configuration data to the ESP32, and the ESP32 can send feedback back to the application. This feedback may include whether the selected drawer was opened, whether it was closed again, or whether the reminder was not completed within the expected time.

5.2. Main Architecture Blocks

The system is divided into five main architecture blocks:

- Mobile application
- ESP32 control unit
- Input subsystem
- Output subsystem
- Physical pill organizer structure

The ESP32 works as the connection point between the software and hardware parts of the system. It receives the configuration from the mobile application, processes the reminder logic, reads the input signals from the drawers, controls the output devices, and sends feedback back to the application.

5.3. General System Flow

The general flow starts when the caregiver configures a medication reminder from the mobile application. The application sends the required information to the ESP32, including the medicine name, drawer number, and scheduled time.

When the scheduled time arrives, the ESP32 activates reminder mode. During this mode, the system provides visual, audible, and textual indications to guide the user toward the selected drawer.

The ESP32 then waits for the drawer interaction. The selected drawer must be opened and then closed again for the interaction to be considered completed. Once this happens, the ESP32 stops the reminder outputs, records the interaction, and can send feedback to the mobile application.

If the selected drawer is not opened or is not closed again within a defined time, the ESP32 can keep the alert active and send a warning notification to the mobile application.

5.4. Architecture Diagrams

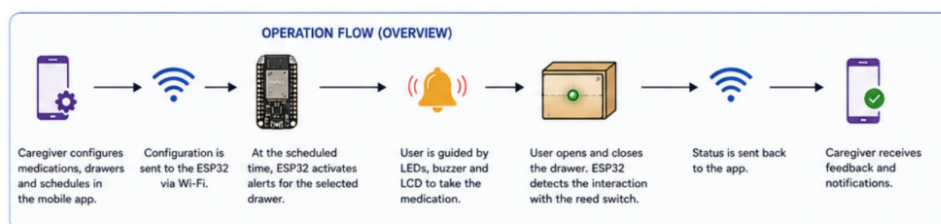


Figure 4: General system architecture of the Smart Pill Reminder Box

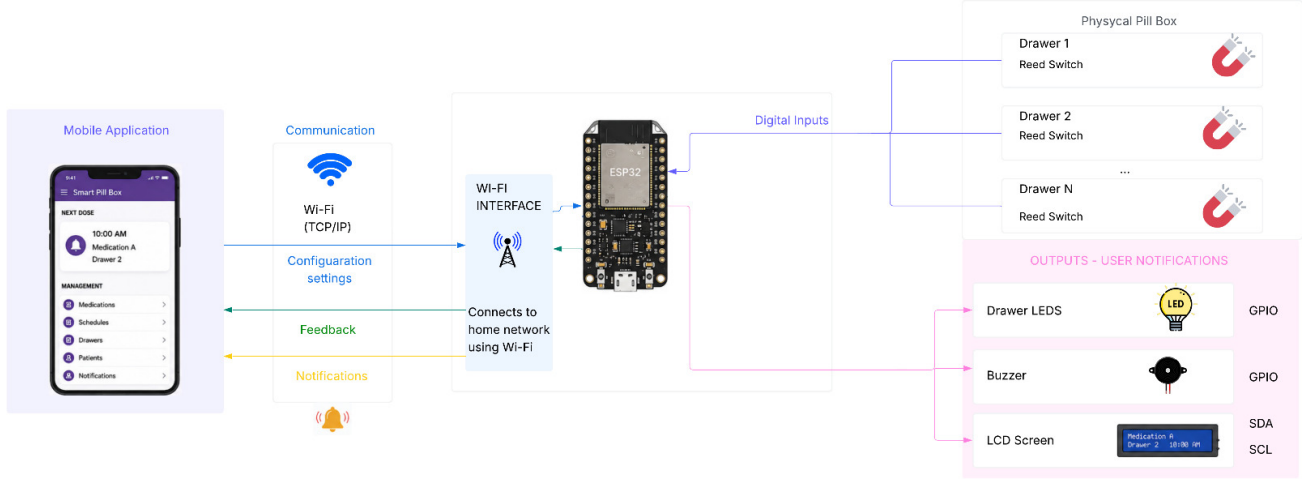


Figure 5: System architecture diagram showing communication flow and component connections

5.5. Hardware/Software Responsibility Split

The mobile application is responsible for configuration and monitoring. It allows the caregiver to define medication information, assign drawers, set schedules, and receive feedback from the system.

The ESP32 is responsible for the control logic of the physical device. It receives configuration data, activates reminders, reads drawer interaction signals, controls the output components, and sends interaction results back to the application.

The physical box provides the structure where the user interacts with the system. It contains the drawers, sensors, actuators, wiring, and electronic support components required for the prototype.

6. Design Decisions

6.1. ESP32 as the Main Microcontroller

The ESP32 was selected as the main microcontroller because it includes integrated WiFi and Bluetooth connectivity, allowing wireless communication between the smart pill organizer and the mobile application. This decision is essential because the system must receive medication schedules from the application and send feedback regarding user interaction with the drawers.

6.2. Integration of a Mobile Application for Caregivers

The system includes a mobile application intended for caregivers or family members, since the primary user may have cognitive impairment and may not always be able to configure the device independently. Through the application, users can register medications, schedules, and assigned drawers. Additionally, the application can send alerts if the patient misses a dose or if the medication supply is running low.

6.3. Use of Illuminated Compartments

Each drawer includes LED lighting to visually indicate which compartment the user must open. This decision improves accessibility and reduces the possibility of mistakes, since the older adult receives not only a general reminder but also a clear and localized indication of the correct medication compartment.

6.4. Combination of Audible and Visual Alarms

The product combines a buzzer, LED lights, and an LCD screen to reinforce reminders through multiple sensory channels. This makes the interaction clearer and more accessible for users who may experience memory, attention, or perception difficulties.

6.5. LCD Screen Integration

The LCD screen is used to display specific information such as the medication name, drawer number, and confirmation messages. This decision allows the user to receive direct textual instructions, reducing confusion during the medication process.

6.6. Use of Magnetic Reed Switches

Magnetic reed switches were selected to detect whether a drawer has been opened and closed. This solution allows the sensors to remain hidden inside the physical structure of the product, improving the aesthetics of the device and avoiding visible mechanical buttons that could confuse the user.

6.7. Drawer Interaction Confirmation

The system does not directly confirm that the user swallowed the medication; instead, it records whether the correct drawer was opened and closed. This is a realistic design decision for a prototype because it verifies interaction with the medication without requiring invasive or overly complex sensors.

6.8. Focus on Autonomy and Independence

The design aims to support older adults without completely removing their participation in the medication routine. Instead of relying entirely on caregivers, the device guides the user step by step, promoting greater independence and reducing the frustration commonly associated with loss of autonomy.

7. Component Distribution

7.1. Mobile Application

The mobile application is located on the software side of the system. Its main function is to allow the caregiver or authorized user to configure the medication reminders. From the application, the caregiver can define the medicine name, the drawer where the medicine is stored, and the scheduled time for the reminder.

The application also receives information from the ESP32. This information can be used to notify the caregiver whether the interaction with the drawer was completed or whether the medicine may not have been taken on time.

7.2. ESP32 Microcontroller

The ESP32 is placed inside the physical pill reminder box and acts as the central controller. It is connected to the sensors and actuators of the system. Its responsibilities include receiving data from the mobile application, checking reminder schedules, reading drawer states, activating alerts, and sending feedback to the application.

The ESP32 must be connected to the magnetic reed switches as digital inputs. It must also be connected to the LEDs, buzzer, and LCD screen as output devices. Because the ESP32 includes wireless communication capabilities, it can communicate with the mobile application without requiring an additional external communication module.

7.3. Magnetic Reed Switches

The magnetic reed switches are distributed across the drawers of the pill organizer. Each drawer has one reed switch installed in the fixed structure of the box and one magnet attached to the moving part of the drawer.

When the drawer is closed, the magnet remains close to the reed switch. When the drawer is opened, the magnet moves away, causing a change in the signal detected by the ESP32. This allows the system to determine whether each drawer is open or closed.

The reed switches are input components because they provide information from the physical box to the ESP32.

7.4. Drawer LEDs

The LEDs are placed around or near each drawer. Their purpose is to visually indicate which drawer has been selected by the system during a reminder.

When a reminder is activated, only the LEDs of the assigned drawer blink. This helps identify the correct compartment and separates the selected drawer from the rest of the box. Once the drawer has been opened and closed again, the ESP32 turns off the LEDs.

The LEDs are output components because they receive control signals from the ESP32.

7.5. Resistors

Resistors are included as supporting electronic components in the physical circuit. They are mainly used to protect the LEDs by limiting the current that flows through them. Each LED or LED group

should include an appropriate current-limiting resistor to prevent excessive current from damaging the LED or the ESP32 output pin.

Resistors may also be used with the magnetic reed switches as pull-up or pull-down resistors, depending on the final wiring configuration. This helps ensure that the ESP32 receives a stable digital signal when a drawer is open or closed.

Therefore, resistors are not user-interaction components, but they are necessary for the correct and safe operation of the electronic circuit.

7.6. Buzzer or Small Speaker

The buzzer or small speaker is installed inside the pill reminder box. It is used to generate the audible alert when a medication reminder is activated.

The buzzer starts sounding when the scheduled time arrives. It remains active during the reminder process and stops only when the selected drawer has been opened and closed again, or when the system logic determines that the reminder has ended.

The buzzer is an output component controlled by the ESP32.

7.7. LCD Screen

The LCD screen is located on the external part of the box, where it can be easily seen by the user. It displays relevant information during the reminder process, such as the medicine name, selected drawer number, instructions, and confirmation or warning messages. Example messages include:

```
Take Medicine A - Drawer 2
Open selected drawer
Close drawer to confirm
Medicine not taken
```

The LCD screen is an output component because it receives display instructions from the ESP32.

7.8. Physical Box and Drawers

The physical box contains the drawers where the medicines are stored. Each drawer is assigned to a specific medication according to the configuration received from the mobile application.

Each drawer includes two main hardware elements: one magnetic reed switch for detecting open and closed states, and one LED indicator or LED group for visual guidance. The structure must allow the sensors to be hidden inside the box while keeping the LEDs visible to the user.

The physical distribution of the components must support both functionality and usability. The sensors must be positioned accurately to detect drawer movement, while the LEDs and LCD screen must be placed in visible locations to guide the user clearly. Figure 5 illustrates how all the described components are interconnected within the system.

8. Used Materials

The Smart Pill Reminder Box will be primarily constructed using transparent acrylic and plywood, selected for their functional, aesthetic, and manufacturing advantages.

The drawers of the device will be made of transparent acrylic. This material was selected because it is affordable, lightweight, and highly compatible with the visual feedback system of the product. Acrylic interacts effectively with light, allowing the LED illumination to spread evenly throughout the drawer and making the selected compartment easier for the user to identify during the reminder process. In addition, acrylic can be precisely cut using laser manufacturing techniques, which results in clean edges, accurate dimensions, and a more polished aesthetic finish that closely matches the technical drawings of the prototype and the original 3D model.

The external structure and remaining components of the system (non-electrical components) will be made of plywood. Plywood was selected because it is inexpensive, lightweight, resistant, and easy to manufacture through laser cutting processes. This facilitates rapid prototyping and allows the physical structure to maintain precise measurements and assembly accuracy. Beyond its functional advantages, wood also contributes positively to the user experience. Wood possesses visual, tactile, and olfactory qualities capable of generating warmth, trust, and well-being. Its organic texture and natural appearance promote an emotional connection between the user and the product, creating a sense of familiarity and comfort. Additionally, wood naturally dampens sound and regulates surface temperature, helping reduce excessive sensory stimulation and supporting a calmer and more accessible interaction experience for older adults.

Another important reason for selecting plywood is that wood is not translucent, which makes it highly effective for concealing and isolating the internal electronic components and wiring of the system from the user. This is especially important in products intended for older adults and patients with neurocognitive conditions, since exposed circuits, cables, or electronic elements may create confusion, anxiety, or unsafe interactions. By fully enclosing the electrical system inside a solid wooden structure, the product provides a safer and less visually overwhelming experience while also protecting the user from accidental contact with electronic components. This design decision improves both the physical safety and the perceived reliability of the device, contributing to a more intuitive and trustworthy interaction for vulnerable users.

Material	Applied To	Key Properties	Manufacturing
Transparent Acrylic	Drawer compartments	Lightweight, affordable, high light transmittance for LED diffusion, smooth finish	Laser cutting
Plywood	External box structure and non-electrical parts	Inexpensive, resistant, opaque (conceals wiring), warm tactile and visual qualities for older adult users	Laser cutting

Table 1: Physical materials selected for the Smart Pill Reminder Box prototype

9. Conclusions

The Smart Pill Reminder Box represents a concrete and technically grounded response to a real healthcare challenge faced by older adults with cognitive impairment. Through the design process documented in this report, the system evolved from an identified problem into a structured hardware and software proposal supported by use case modeling, class design, state-based behavior, and deliberate material and component selection.

The use of an ESP32 microcontroller as the central control unit proved to be a well-justified decision, as its integrated wireless capabilities eliminate the need for additional communication hardware while providing sufficient GPIO resources to manage all input and output components of the prototype. The combination of magnetic reed switches, drawer LEDs, a buzzer, and an LCD screen creates a multimodal reminder experience that addresses the sensory and cognitive needs of the target user without requiring complex interaction from them.

The object-oriented class design, organized across four clearly defined layers — mobile application, domain model, device services, and hardware abstraction — provides a maintainable and extensible architecture. The use of the MQTT protocol for bidirectional communication between the mobile application and the ESP32 ensures a lightweight and reliable data exchange suitable for a home-use IoT device.

The material selection reinforces both the functional and experiential goals of the product. Transparent acrylic maximizes the effectiveness of the LED visual feedback system, while plywood provides a warm, familiar, and safe enclosure that conceals all electronics from the user and contributes positively to the interaction experience of older adults.

Overall, this design document establishes a solid foundation for the physical construction and firmware development phases of the prototype. The decisions made throughout this process prioritize accessibility, safety, and user autonomy, ensuring that the device serves not only as a medication reminder, but as a tool that supports dignity and independence in the daily lives of its users.

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